Supplementary Material: Risk-controlled post-hoc rescue of rare cell types in single-cell RNA-seq under label scarcity

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This document is the standalone Supplementary Material for the main manuscript *"Risk-controlled post-hoc rescue of rare cell types in single-cell RNA-seq under label scarcity"*. Section, table and figure numbers here are prefixed with S and are self-contained; cross-references to the main text (e.g. Fig. 1, §3) refer to the main manuscript.

S1 Supplementary material

S1. Benchmark and reproduction protocol

All experiments use six datasets, nine methods, four labelled fractions $(0.01, 0.05, 0.10, \text{all})$ and three seeds $(42, 43, 44)$ under a donor-held-out $70/15/15$ split. scRareRefine consumes frozen scANVI latents and predictions; prototypes are estimated on labelled training cells and all thresholds on the validation partition. The FFR budget is fixed at $\alpha = 0.01$ for every dataset and is never tuned. Split hashes and per-run manifests are released with the code.

S2. Significance, scRareRefine vs. each method

The 54–72 paired cells in our benchmark are not independent samples: within a dataset, seeds share the same donor split up to resampling noise, and (§S3) several nominal fractions collapse to an identical labelled set. A per-cell test therefore overstates the effective sample size. We report two analyses. Table S1 is the primary, conservative one: a *dataset-clustered bootstrap* that resamples whole datasets ($20\,000$ resamples with replacement from the six biologically independent datasets, pooling all seeds/fractions within a resampled dataset), together with a dataset-level sign test on the 6 per-dataset mean ΔF1 values (win = scRareRefine's per-dataset mean F1 exceeds the comparator's). Table S2 is the secondary, liberal per-configuration analysis used in earlier drafts of this benchmark (each of the $54/72$ dataset×fraction×seed cells treated as an independent pair), which we retain for continuity but whose p-values should not be read as exact. The two analyses agree in direction and effect size for every comparator; the dataset-clustered CIs are wider, as expected once the clustering is accounted for.

The lone dataset-level tie against scANVI in both regimes is the small-intestine tuft dataset, where the backbone is already near ceiling and scRareRefine correctly abstains rather than risking a false rescue; the sign-test $p = 0.031$ is thus the exact one-sided probability of ≥ 5 wins out of 5 non-tied datasets, and is the more defensible headline number for this benchmark's true unit of replication.

S3. Distinct labelled configurations

Because the labelled rare budget is $\max(5, \lfloor p\, N^r_{\text{tr}} \rfloor)$, small-rare datasets reach the five-cell floor at more than one nominal fraction, collapsing the 18 nominal scarce configurations to 15 distinct ones. On Baron pancreas the 0.01 and 0.05 fractions coincide (5 labelled cells); on stomach mast cells all three scarce

Table S1: Dataset-clustered bootstrap comparison of scRareRefine against each method (primary test). Whole datasets ($n = 6$) are resampled with replacement, 20 000 times. W/T/L counts compare per-dataset mean F1 (ties are exact equality, to within floating-point precision). Full benchmark and label-scarce regime. † HiCat is transductive (upper-bound reference).

Regime	Comparator	W/T/L (datasets)	mean ΔF1	Bootstrap 95% CI	Boot. $p(\Delta \le 0)$	Sign-test p
Full	scANVI	5/1/0	$+0.13$	$[0.06, 0.19]$	1.0×10^{-4}	0.031
	k-NN	6/0/0	$+0.12$	$[0.03, 0.24]$	$< 5 \times 10^{-5}$	0.016
	CellTypist	6/0/0	$+0.18$	$[0.08, 0.27]$	$< 5 \times 10^{-5}$	0.016
	scBalance	6/0/0	$+0.17$	$[0.08, 0.27]$	$< 5 \times 10^{-5}$	0.016
	ProtoCloud	6/0/0	$+0.16$	$[0.07, 0.26]$	$< 5 \times 10^{-5}$	0.016
	scCAD	6/0/0	$+0.33$	$[0.22, 0.42]$	$< 5 \times 10^{-5}$	0.016
	TOSICA	6/0/0	$+0.37$	$[0.20, 0.54]$	$< 5 \times 10^{-5}$	0.016
	HiCat†	6/0/0	$+0.52$	$[0.42, 0.63]$	$< 5 \times 10^{-5}$	0.016
Scarce	scANVI	5/1/0	$+0.16$	$[0.08, 0.25]$	1.0×10^{-4}	0.031
	k-NN	6/0/0	$+0.15$	$[0.05, 0.29]$	$< 5 \times 10^{-5}$	0.016
	CellTypist	6/0/0	$+0.25$	$[0.12, 0.37]$	$< 5 \times 10^{-5}$	0.016
	scBalance	6/0/0	$+0.23$	$[0.11, 0.36]$	$< 5 \times 10^{-5}$	0.016
	ProtoCloud	6/0/0	$+0.23$	$[0.09, 0.36]$	$< 5 \times 10^{-5}$	0.016
	scCAD	6/0/0	$+0.35$	$[0.24, 0.44]$	$< 5 \times 10^{-5}$	0.016
	TOSICA	6/0/0	$+0.39$	$[0.22, 0.54]$	$< 5 \times 10^{-5}$	0.016
	HiCat†	6/0/0	$+0.69$	$[0.57, 0.82]$	$< 5 \times 10^{-5}$	0.016

fractions coincide (5 cells). Over the 15 distinct configurations scRareRefine beats the majority of the eight comparators in $15/15$ and is the single best method in $14/15$ (the exception is the fully saturated small-intestine tuft setting at $p = 0.10$, where it ties the best baseline). Table S3 lists the distinct configurations and the per-configuration mean F1.

S4. Separability-gate threshold sensitivity

A benchmark-wide sweep of the gate threshold $\rho_{\text{low}} \in \{0.0, 0.7, 1.0, 1.3, 1.6\}$ over the label-scarce regime ($n = 54$): at $\rho_{\text{low}} \le 1.0$ mean F1 is 0.902 with two configurations exceeding the FFR budget (worst-case 0.0153); at $\rho_{\text{low}} = 1.3$ mean F1 is 0.878 with zero violations (worst-case 0.0098); at $\rho_{\text{low}} = 1.6$ mean F1 falls to 0.838 with no further FFR benefit. The value 1.3 is thus the smallest threshold that keeps every configuration within budget, and is fixed across all datasets.

S5–S8. Further material

The released repository additionally provides: robustness curves over the labelled fraction (S5); runtime and peak memory per method (S6); additional qualitative UMAP rescue panels for stomach and Baron pancreas (S7); and full provenance — split hashes, code versions, and per-run manifests (S8).

Table S2: Secondary, per-configuration analysis (liberal; retained for continuity with earlier drafts). Full benchmark ($n = 72$) and label-scarce regime ($n = 54$) dataset×fraction×seed cells treated as independent pairs. † HiCat is transductive (upper-bound reference).

Regime	Comparator	W/T/L	mean ΔF1	95% CI	Wilcoxon p
Full	scANVI	34/37/1	$+0.13$	$[0.07, 0.19]$	1.6×10^{-7}
	k-NN	55/9/8	$+0.12$	$[0.09, 0.16]$	2.1×10^{-10}
	CellTypist	54/2/16	$+0.18$	$[0.12, 0.23]$	3.1×10^{-8}
	scBalance	53/7/12	$+0.17$	$[0.12, 0.23]$	1.0×10^{-8}
	ProtoCloud	52/5/15	$+0.16$	$[0.11, 0.22]$	4.0×10^{-8}
	scCAD	68/2/2	$+0.33$	$[0.28, 0.38]$	2.0×10^{-13}
	TOSICA	71/1/0	$+0.37$	$[0.31, 0.43]$	1.2×10^{-13}
	HiCat†	61/2/9	$+0.52$	$[0.43, 0.61]$	6.5×10^{-12}
Scarce	scANVI	29/25/0	$+0.16$	$[0.09, 0.24]$	1.3×10^{-6}
	k-NN	46/6/2	$+0.15$	$[0.11, 0.20]$	1.9×10^{-9}
	CellTypist	51/1/2	$+0.25$	$[0.19, 0.32]$	1.8×10^{-10}
	scBalance	47/5/2	$+0.23$	$[0.17, 0.30]$	9.7×10^{-10}
	ProtoCloud	48/4/2	$+0.23$	$[0.17, 0.29]$	5.1×10^{-10}
	scCAD	52/1/1	$+0.35$	$[0.29, 0.41]$	1.3×10^{-10}
	TOSICA	53/1/0	$+0.39$	$[0.32, 0.45]$	1.2×10^{-10}
	HiCat†	52/1/1	$+0.69$	$[0.61, 0.77]$	1.3×10^{-10}

Table S3: Distinct labelled configurations in the label-scarce regime and scRareRefine mean rare-cell F1 (three seeds). Collapsed fractions share a labelled set and are listed once.

Dataset (rare type)	Fraction(s)	# labelled rare	scRareRefine F1
immune dendritic (ASDC)	0.01	5	0.93
immune dendritic (ASDC)	0.05	13	0.94
immune dendritic (ASDC)	0.10	27	0.94
Baron pancreas (gamma)	0.01, 0.05	5	0.57
Baron pancreas (gamma)	0.10	10	0.84
integrated pancreas (endo.)	0.01	5	1.00
integrated pancreas (endo.)	0.05	12	1.00
integrated pancreas (endo.)	0.10	24	0.99
lung (lymphatic EC)	0.01	5	0.98
lung (lymphatic EC)	0.05	10	0.98
lung (lymphatic EC)	0.10	20	0.97
stomach (mast)	0.01,0.05,0.10	5	0.72
small intestine (tuft)	0.01	5	0.98
small intestine (tuft)	0.05	15	0.98
small intestine (tuft)	0.10	31	0.98